

MONDY, Russian Federation

WMO: 308020

Lat: 51.683N

Lon: 100.983E

Elev: 4279

StdP: 12.56

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.2	-26.6	-34.7	0.9	-30.6	-30.0	1.2	-25.5	19.0	7.4	17.0	10.3	0.4	290	0.696

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.1	77.7	58.5	74.3	57.6	71.1	56.6	62.1	71.7	60.1	69.3	58.5	67.4	6.4	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
58.5	85.9	66.5	56.6	80.0	63.8	54.8	75.0	62.1	30.1	71.7	28.6	69.0	27.3	67.5	70.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
17.5	15.1	13.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	29.4	-1.6	1.6	16.7	31.4	42.8	54.5	59.6	54.7	43.1	30.2	14.1	3.3
	DBStd	22.83	9.93	10.80	10.78	8.89	7.28	6.44	4.58	5.18	6.31	8.55	10.78	10.76
	HDD50	8202	1601	1356	1032	558	244	30	1	20	222	613	1078	1447
	HDD65	13021	2066	1776	1497	1007	689	317	175	318	658	1078	1528	1912
	CDD50	664	0	0	0	1	19	165	298	167	14	0	0	0
	CDD65	9	0	0	0	0	0	2	7	0	0	0	0	0
	CDH74	317	0	0	0	0	18	103	164	30	2	0	0	0
	CDH80	37	0	0	0	0	2	14	21	0	0	0	0	0
Wind	WSAvg	4.7	4.0	4.1	5.4	6.0	6.1	4.6	3.7	3.8	4.6	4.5	4.6	4.5
Precipitation	PrecAvg	13.1	0.1	0.1	0.2	0.4	0.9	2.5	4.0	3.0	1.3	0.3	0.2	0.1
	PrecMax	18.5	0.2	0.3	0.4	0.9	1.9	5.3	7.6	4.6	2.8	1.0	0.5	0.5
	PrecMin	8.7	0.0	0.0	0.0	0.1	0.2	0.6	2.2	0.9	0.2	0.1	0.0	0.0
	PrecStd	2.8	0.0	0.1	0.1	0.2	0.4	1.0	1.2	1.0	0.7	0.2	0.1	0.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	28.8	35.5	51.0	64.1	77.2	82.4	82.8	77.4	71.2	59.0	46.4	31.2
		MCWB	23.2	26.7	36.7	45.8	52.5	59.4	63.2	57.5	52.1	42.6	36.2	24.7
	2%	DB	22.3	29.4	45.9	57.0	69.6	77.2	78.2	73.5	65.3	53.8	40.1	26.8
		MCWB	17.6	22.7	34.1	42.1	49.7	56.5	60.6	58.2	50.8	39.8	31.7	21.4
	5%	DB	17.8	24.7	41.0	52.5	64.8	72.9	74.8	70.2	60.7	49.3	35.2	23.5
		MCWB	14.1	19.2	30.7	38.8	47.4	56.1	59.7	56.9	47.9	38.2	28.6	18.9
	10%	DB	13.5	19.9	36.1	48.5	60.0	69.2	71.2	66.6	56.1	45.6	30.8	19.9
		MCWB	10.5	15.7	27.6	36.5	44.7	54.7	58.4	55.7	45.9	36.3	25.3	16.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.5	27.3	37.5	46.6	54.8	62.5	66.3	62.3	56.0	44.2	36.5	25.0
		MCDB	28.7	35.2	50.0	60.8	73.2	75.5	78.9	72.4	65.5	54.9	45.4	30.2
	2%	WB	17.8	22.9	34.3	42.8	51.2	59.3	63.3	60.1	52.8	41.0	31.9	21.5
		MCDB	22.0	29.3	45.6	55.8	67.1	70.9	73.4	69.9	61.2	51.2	39.5	26.2
	5%	WB	14.1	19.4	31.1	39.4	48.4	57.5	61.4	58.4	50.3	38.8	28.9	19.1
		MCDB	17.6	24.5	40.5	50.5	62.2	68.4	70.0	66.8	58.0	48.5	35.0	23.4
	10%	WB	10.7	15.8	28.0	37.0	45.9	55.8	59.7	56.6	47.1	36.5	25.6	16.2
		MCDB	13.5	20.0	35.8	47.7	58.0	67.0	68.3	64.0	54.9	44.8	30.4	19.8
Mean Daily Temperature Range	5% DB	MDBR	24.0	29.3	28.6	22.8	22.7	21.4	20.1	19.7	22.0	23.8	22.9	21.4
		MCDBR	25.9	31.8	33.7	31.1	33.9	31.7	28.1	28.4	32.2	32.0	25.3	21.5
		MCWBR	22.1	25.7	23.1	18.4	17.2	14.4	12.5	13.6	17.2	19.2	17.8	17.6
	5% WB	MCDBR	25.8	31.7	31.4	28.5	28.9	25.3	22.6	22.5	26.3	30.0	24.9	21.5
		MCWBR	22.0	25.6	21.5	17.7	16.1	12.7	12.0	12.0	15.0	18.5	17.6	17.7
Clear-Sky Solar Irradiance	taub	0.232	0.258	0.301	0.347	0.379	0.373	0.369	0.343	0.303	0.281	0.253	0.232	
	taud	2.237	2.200	2.140	2.195	2.200	2.346	2.400	2.447	2.501	2.427	2.270	2.193	
	Ebn at Noon	254	276	282	281	277	279	279	281	283	267	240	230	
	Edh at Noon	21	29	38	41	43	37	35	32	27	23	20	18	
All-Sky Solar Radiation	RadAvg	421	736	1196	1630	1877	1805	1682	1483	1166	771	440	315	
	RadStd	15	24	53	71	103	121	122	85	80	44	25	15	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-326	N/A	N/A

Nomenclature: See separate page